

Topics in Multiagent Learning

MIT 6.7980 — Fall 2026

Lecture: Tuesdays and Thursdays, 11:00am-12:30pm, **in room E25-111**.

Instructors:

- ▶ Prof. Gabriele Farina, office 45-501F (in the College of Computing building)
✉ gfarina@mit.edu URL: www.mit.edu/~gfarina
- ▶ Prof. Constantinos Daskalakis, office 32-G694 (in the Stata building)
✉ costis@csail.mit.edu URL: people.csail.mit.edu/costis

We are happy to meet with students by appointment.

Teaching assistants:

- ▶ Kat Federova (✉ fedorova@mit.edu). Office hours: TBD.
- ▶ Mingyang Liu (✉ liumy19@mit.edu). Office hours: TBD.
- ▶ Daniel Xia (✉ dxia03@mit.edu). Office hours: TBD.
- ▶ Rui Yao (✉ rayyao@mit.edu). Office hours: TBD.

Grading:

- ▶ Attendance and participation (see below) — 20%.
- ▶ Improving material (see below) — 30%.
- ▶ Project (see below) — 50%.

Attendance: We expect everyone to attend at least 50% of the lectures. The “Attendance and participation” component of grading above reflects (in a binary manner) whether that was met. We will measure attendance using random quizzes during classes.

Coursework: There are no assigned homework sets. Students will contribute to the shared course materials and complete a project, as described below.

Prerequisites: Discrete Mathematics and Algorithms at the advanced undergraduate level; mathematical maturity.

Lecture notes: Lecture notes are available as HTML and PDF on the course website. Announcements and administrative materials will be posted on Canvas.

Collaboration policy: We encourage working together on the course materials, projects, and discussion of the ideas. Contributions and project work should reflect *your own* understanding. Acknowledge collaborators and sources, and do not present somebody else’s work as your own.

Acceptable AI use: ¹ Students may use generative AI tools (e.g., LLMs) to support learning, brainstorming, editing, debugging, or generating explanations. We believe that *your human understanding* is still the ultimate goal of education more broadly. We should aim to expand our knowledge, deepen our understanding, and sharpen our thinking. Generative AI might be a great tool for aiding this process, but learning and growth require productive struggle. Don’t let generative AI remove or reduce your productive struggle. We also expect (and require) *full transparency* regarding your use of AI. Always indicate if and how you used generative AI, i.e., which tools you used and during which phases of your work.

¹This policy is inspired *in part* from the [template policy](#) provided by the MIT Teaching + Learning Lab.

Description

This course studies multiagent systems through game theory, optimization, and learning theory. We cover foundational topics such as Nash equilibria, regret minimization, learning dynamics, and extensive-form games.

We also explore modern topics: multiagent deep reinforcement learning; information and mechanism design; team games and hidden-role games; alignment; high-dimensional and kernelized learning; nonconvex games; calibration; and the complexity of finding equilibria. Applications and open research questions connect the theory to multiagent AI.

Improving Material

We would like to make the lecture notes available to as many people as possible. You can now read them in a browser, follow links between sections and references, and move between the notes and their source. We would like everyone's help to make this a useful resource for learners around the world.

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Topics in Multiagent Learning

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We also explore modern topics, including multiagent deep reinforcement learning, calibration and alignment, and hidden-role games, connecting theory to applications and open research questions in multiagent AI.

[Schedule & notes](#) [Syllabus \(PDF\)](#) [Fog of War Chess challenge](#)

MIT 6.7980
Topics in Multiagent Learning
Comparative Analysis and Game Theory

LESSONS

1. Solving and equilibria in the Nash equilibrium
2. Extensive games: Nash, Bayesian, and beyond

3. Properties and solutions of the Nash equilibrium
4. Learning algorithms in games
5. Branch and bound
6. Properties of extensive form games
7. Learning in extensive form games
8. Fictitious play and Nash equilibria

THIS LECTURE

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It is straightforward to verify that ϕ is well-defined and maps strategy profiles into strategy profiles. Indeed, the numerator in 1 is always nonnegative, and the denominator is always at least 1, implying that $\phi_{i,s} \geq 0$ for all $s_i \in A_i$ and player $i \in [n]$. Furthermore,

$$\forall i \in [n], \sum_{s_i \in A_i} \phi_{i,s_i}(\pi_{-i}) = \sum_{s_i \in A_i} \frac{\pi_{i,s_i} + [\pi_{i,s_i}(\pi_{-i}, s_{-i})]}{1 + \sum_{s_i \in A_i} \pi_{i,s_i}(\pi_{-i}, s_{-i})} = 1,$$

where we used the fact that $\sum_{s_i \in A_i} \pi_{i,s_i} = 1$ since π_i is a valid strategy. Finally, observe that ϕ is a continuous function. The following example visualizes the Nash improvement function in the small games we have seen so far.

Example 13.7. The plot below visualizes the displacement $\phi(s_1, s_2) - (s_1, s_2)$ induced by the Nash improvement function for four games, whose payoff matrices are noted below each plot, after projecting onto the probability of the first action of each player (and keeping around only the probability of the second action, which is sufficient to uniquely recover the strategy of the player since each player only has two actions). The black dots denote the first points of the Nash improvement function. These correspond exactly to the Nash equilibria of the game, as we make formal below.

The background of the plots highlights the angle of displacement induced by the Nash improvement function, according to the gradient below shown here.

[If the color scheme seems arbitrary, as a small spoiler it will play a fundamental role in the proof of the computational complexity of Nash equilibria, which we will discuss later on in this course. In particular, the three regions of the coloring scheme will be key in defining an important combinatorial construction called *spencer coloring*.]

Example L7.1. As a running example, we will illustrate the representation by analyzing the game tree of a simplified two-player variant of poker (perhaps the archetype of imperfect-information extensive-form games), known as *Kuhn poker* [Kuh50]. As noted by Kuhn himself, even the previous small game already captures certain aspects of deceptive behavior such as *bluffing* and *underbidding*.

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We will divide the class into groups, each focusing on a different part of the material. Using the class GitHub repository, each group can open issues to identify improvements and submit pull requests to implement them. We will improve the material together, reviewing and building on one another's contributions.

Contributions can include clarifying explanations and proofs, fixing errors, adding examples and homework-style exercises for future readers, and polishing figures, organization, and presentation. If anyone is brave enough, we would also love interactive components that let readers experiment with the ideas.

On the bright side, there is no homework! :-) Improving the shared material accounts for 30% of the course grade.

Project

Projects may be completed individually or in groups of 2-3 students and will include a presentation. We will offer three project directions:

Fog of War Challenge. Build and evaluate an agent that plays with partial information. Explore how it uses observations, reasons about uncertainty, and chooses strategic actions. The rules, starter code, and arena are maintained in the separate challenge repository.

Modeling questions. Formulate a multiagent problem by specifying the players, objectives, information, and available actions. Study how modeling choices affect the resulting strategic behavior.

Theory questions. Investigate a mathematical question about equilibria, learning dynamics, or computational complexity. Develop rigorous proofs, bounds, or counterexamples that clarify the behavior of multiagent systems.

The project is the central component of the course and accounts for 50% of the final grade. We will therefore be “robust” in our grading: we will look carefully at the depth of your understanding, the quality and substance of your work, and how clearly you explain your results.

Schedule

0	Sep 10 th	Course Overview We will discuss the syllabus, projects, administrative details, and an overview of the topics covered.	<i>Constantinos Daskalakis; Gabriele Farina</i>
PART I: FOUNDATIONS (8 LECTURES)			
1	Sep 15 th	Setting and equilibria: the Nash equilibrium Nash's existence theorem and its connection to fixed-point theorems.	<i>Constantinos Daskalakis</i>
2	Sep 17 th	Brouwer and Sperner Sperner's lemma, Brouwer's theorem, and combinatorial proofs of equilibrium existence.	<i>Constantinos Daskalakis</i>
3	Sep 22 nd	Properties of Nash equilibrium Topological and computational properties. Zero-sum games and linear programming. Correlated and coarse correlated equilibria.	<i>Gabriele Farina</i>
4	Sep 24 th	Learning in games: Foundations Regret and hindsight rationality. Regret minimization and its relationships with equilibrium concepts.	<i>Gabriele Farina</i>
5	Sep 29 th	Learning in games: Algorithms General principles for learning algorithms. Follow-the-leader, regret matching, multiplicative weights, and online mirror descent.	<i>Gabriele Farina</i>
6	Oct 1 st	Learning with bandit feedback Partial feedback and exploration. From multiplicative weights to Exp3; regret guarantees.	<i>Constantinos Daskalakis</i>
7	Oct 6 th	Modeling extensive-form games Perfect and imperfect information. Kuhn's theorem. Normal-form and sequence-form strategies.	<i>Gabriele Farina</i>
8	Oct 8 th	Learning in extensive-form games No-regret learning, counterfactual utilities, and counterfactual regret minimization (CFR).	<i>Gabriele Farina</i>
	Oct 13 th	No class MIT follows a Monday schedule.	
9	Oct 15 th	Taking stock We will discuss big open questions in the field and possible project ideas.	<i>Gabriele Farina; Constantinos Daskalakis</i>
PART II: MULTIAGENT DEEP REINFORCEMENT LEARNING (2 LECTURES)			
10	Oct 20 th	Multiagent deep RL (part I) Reinforcement learning in games. Self-play and deep learning methods for perfect-information games.	<i>Gabriele Farina</i>
11	Oct 22 nd	Multiagent deep RL (part II) Deep reinforcement learning for imperfect-information games. Game-playing applications, including Generals.io.	<i>Gabriele Farina</i>

PART III: INFORMATION, COMMUNICATION, ALIGNMENT (4 LECTURES)		
12	Oct 27 th	Information and mechanism design Designing information and incentives in strategic interactions. <i>Brian Hu Zhang</i>
13	Oct 29 th	Team games and hidden-role games Coordination in teams and games with hidden roles. <i>Brian Hu Zhang</i>
14	Nov 3 rd	Alignment (part I) Reinforcement learning from human feedback (RLHF) and alignment. <i>Natalie Collina</i>
15	Nov 5 th	Alignment (part II) Regularized RLHF and direct preference optimization (DPO). <i>Natalie Collina</i>
PART IV: ADVANCED LEARNING (3 LECTURES)		
16	Nov 10 th	Nonconvex games Nonconvexity, learning dynamics, and local equilibrium concepts. <i>Weiqiang Zheng</i>
17	Nov 12 th	High-dimensional and kernelized learning Learning with large strategy spaces. Kernelized methods and multiplicative weights. <i>Constantinos Daskalakis</i>
18	Nov 17 th	Calibration Calibrated prediction and its connections to learning in games. <i>Constantinos Daskalakis</i>
PART V: COMPUTATIONAL COMPLEXITY (2 LECTURES)		
19	Nov 19 th	Total search and TFNP Total search problems, the TFNP framework, and the PPAD complexity class. <i>Constantinos Daskalakis</i>
20	Nov 24 th	PPAD-hardness of Nash equilibrium Reductions and the computational hardness of finding Nash equilibria. <i>Constantinos Daskalakis</i>
	Nov 26 th	No class Thanksgiving holiday.
PROJECT WORK AND PRESENTATIONS		
21	Dec 1 st	Project work Time reserved for project development and preparation.
22	Dec 3 rd	Project work Time reserved for project development and preparation.
23	Dec 8 th	Project presentations
24	Dec 10 th	Project presentations